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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Mixed Signals
Selected Domain	Domain 2 — Orbyte Smart SME Challenge: Less Busywork, Better Business
Selected SDG Track (1–17)	SDG 8 - Decent Work and Economic Growth
Project / Solution Name	Glacier Link
Name (Leader)	Soh Ning Xuan
Email & Phone (Leader)	ninxuannnn06@gmail.com, 012-6370306
University Affiliation(s)	University of Nottingham Malaysia, University of Southampton Malaysia, Monash University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small food businesses such as bakeries, cafés, frozen-food sellers and mini-marts depend on chillers and freezers to protect perishable stock. When a unit fails or drifts outside its safe temperature range, staff may notice late, report the fault without enough detail, or be unsure what to do next. Owners then have to identify the affected unit, estimate stock at risk, decide whether stock can be moved, check warranty coverage, choose a vendor and obtain any required approval. These steps are often coordinated through calls and messages, making it difficult to see who is responding and whether the temperature has recovered.

The consequences include potential food disposal, lost sales, disrupted service, staff stress and time spent coordinating repairs. Spoiled food also wastes the water, energy, land and other resources used to produce and transport it. Small operators may have limited time and working capital to absorb a refrigeration incident. Gracier Link addresses this response gap by helping teams act promptly and keep a traceable record from the first report through stock protection, repair coordination and recovery verification. Global food-waste data provides context, but local pilot data is needed to quantify refrigeration-related losses among the target businesses.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Gracier Link is a web application that guides staff and managers through a refrigeration incident. Staff can submit a plain-language fault report or record a temperature reading. The prototype identifies the likely unit and symptom, applies configured safe-temperature limits and presents an incident with severity, stock-at-risk context and recommended next steps. If a report is ambiguous, the workflow can request clarification.

The current application uses deterministic rules for temperature severity, permissions, approval limits and workflow transitions. An optional Claude parser can interpret report text when explicitly enabled; it is disabled by default and has a rules-based fallback. AI does not decide temperature severity, repair approval or incident recovery. Staff can record a stock transfer, authorized managers can approve or reject eligible vendor costs, and users can track dispatch and repair progress. A fresh in-range reading is required to verify recovery.

Role-based access and an audit timeline make actions visible. Demo mode uses fictional businesses, simulated time and mock vendor messages; it does not contact vendors or prove real-world savings. The Phase 2 plan is to test the workflow with small-business users, refine report clarification and prepare a controlled deployment. The team intends to apply workshop learning in GenAI integration, database design and cloud hosting while keeping safety and approval decisions explicit and reviewable.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React, TypeScript, Vite and Tailwind CSS. Backend: Python 3.12, FastAPI and SQLModel, exposed through REST endpoints. Database: SQLite for the current prototype. Authentication: Clerk in normal mode; demo mode simulates users and roles. The optional Anthropic API parser is disabled by default; deterministic rules handle parsing when it is off or fails. Notifications are recorded in an internal mock WhatsApp outbox and are not externally delivered. The repository documents a Replit deployment path, but a live deployment and sponsor-provided service use were not confirmed.

System flow: Staff/manager browser → React UI → FastAPI API → SQLModel/SQLite. The API invokes cold-chain rules for readings, permissions, approvals and recovery; optional Claude text parsing can assist report interpretation when enabled. Audit events and mock notifications are stored by the application.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The workflow is built around three roles: staff who report faults and protect stock, supervisors or owners who approve costs within their limits, and managers who track vendors and verify recovery. Each incident shows its status, the recommended next action and an audit timeline. Role checks and approval thresholds keep decisions with authorised users. Recovery is not recorded while the latest reading is still out of range, so a completion tap alone is not proof that storage is safe again.

For Phase 2, we will run scenario-based usability tests instead of field interviews. Three to five participants outside the team will play staff, supervisor and owner on their phones through scripted incidents: a freezer failure, a door-opening reading that should not alarm, an approval above the supervisor's limit, and a repair completed while the unit is still warm. We will record time to identify the next action, wrong or blocked steps and confusing wording, then fix the most common issues before the final demo. No research findings are claimed yet. Owner interviews, including checking the RM150 limit, are planned for the pilot. Deployment needs a secured hosted database, access controls, backups and a retention policy.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.2: "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors." SDG 8.3: "…encourage the formalization and growth of micro-, small- and medium-sized enterprises…" Glacier Link is a technology upgrade for labour-intensive micro and small food businesses: it automates the detection, decision and repair-coordination work that owners and staff currently do by hand.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Malaysia's MSMEs employ 8.09 million people (48.7% of the workforce) and generate 39.7% of GDP, yet labour productivity sits at RM85,299 per person (DOSM, 2026). In a small bakery or café, a failing freezer drags that down further: the owner, often at home in the evening, has to work through about 8 manual steps to get it fixed. They need to work out which unit it is, check what's inside, check the warranty, find a technician, call, and keep chasing (team estimate from mapping the current process). We'll time how long this takes in scenario tests as our measured baseline.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Primary KPI: median minutes from fault detection to technician booked per incident. The prototype already records this automatically (median_min_detect_to_dispatch in /v1/metrics). Secondary KPIs: (1) manual owner/manager actions per incident; (2) share of call-out decisions made by the correctly authorised role, with no unauthorised approvals.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	In timed scenario tests with 3–5 participants using the same synthetic freezer-failure incident: ≥ 50% reduction in median detection-to-booking time versus the same participants handling it manually (phone/WhatsApp), and owner/manager actions reduced from about 8 to 1 (one approval tap), an ~87% reduction, with 100% of call-outs decided by the authorised role. These are proposed targets measured in simulated conditions, not achieved results or field data. We will report participant count, raw times for both methods and the percentage change. A field pilot at real outlets would follow Phase 2.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☒ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☒ Yes ☐ No [CONFIRM dates and ownership: repository/prototype materials appear to pre-date this submission.]

☒ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☒ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

Pre-existing work: Gracier Link repository, application code, interface, design/planning documents and synthetic demo scenario, as applicable. Team members and creation dates must be identified by the team before signing.

☒ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☑ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Date: 24/10/2026